Fish electrofishing, gill netting, and fyke netting counts (DP1.20107.001)

Measurement

Fish taxonomy, abundance, and physical characteristics of fish in wadeable streams and lakes. Physical characteristics recorded include: length, width, life stage, deformities, eroded fins, lesions, tumors, and parasites. Vouchers and tissue samples are collected as needed.

Collection methodology

Each NEON aquatic stream site is divided into 10 designated reaches measuring between 80m and 120m. Of these 10 reaches, 3 are "fixed" meaning they are sampled twice every year. Fixed reaches are sampled using a three-pass electrofishing depletion strategy. The remaining 7 reaches are rotated through and are sampled using a single pass. Lake sites are similar but instead use a segmented approach to capture nearshore and offshore populations. These segments are located between two riparian transects and they all meet in the center of the lake. The fixed segments of lakes include habitat features that best represent the lake and include one fyke net and one gill net set. At both streams and lakes the same random reaches/segments are sampled in spring and fall and rotate to a new set each year. One full sampling bout will contain the 3 fixed reaches and 3 random reaches for a total of 6.

For information about disturbances, land management activities, and other incidents that may impact data at NEON sites, see the Site management and event reporting (DP1.10111.001) data product.

Data package contents

fsh_bulkCount: Fish sampling field data bulk count
fsh_perFish: Fish sampling field data per specimen captured
fsh_perPass: Fish sampling field data per pass
fsh_fieldData: Fish sampling field data per sub-sampling reach
fsh_invertBycatch: Fish sampling macroinvertebrate bycatch
fsh_morphospecies: Fish morphospecies resolution data
fsh_identificationHistory: Fish identification history for records where identifications have changed
fsh_invertBycatchIDHistory: Macroinvertebrate bycatch identification history for records where identifications have changed
variables: Description and units for each column of data in data tables
readme: Data product description, issue log, and other metadata about the data product
validation: Description of data validation applied at the points of collection and ingest

Data quality

Metadata about conditions and equipment settings during sampling can be found in the fsh_perPass table. In the tables containing taxonomic identifications, the identificationQualifier field indicates uncertainty in the identification, and the identifiedBy field can be used to estimate observer effects.

Please note that quality checks are comprehensive but not exhaustive; therefore, unknown data quality issues may exist. Users are advised to evaluate quality of the data as relevant to the scientific research question being addressed, perform data review and post-processing prior to analysis, and use the data quality information and issue logs included in download packages to aid interpretation.

Standard calculations

For wrapper functions to download data from the API, and functions to merge tabular data files across sites and months, NEON provides the neonUtilities package in R and the neonutilities package in Python. See the Download and Explore NEON Data tutorial for introductory instructions in both programming languages.

Fish capture data are stored across multiple interconnected tables that separately track individual fish measurements, bulk species counts, and details about sampling effort. The Introduction to NEON Fish Capture Data tutorial demonstrates how to properly join and summarize tables in order to calculate total species-by-pass capture counts.

Table joining

Table 1	Table 2	Join by field(s)
fsh_fieldData	fsh_perPass	reachID
fsh_perFish	fsh_perPass	eventID
fsh_bulkCount	fsh_perPass	eventID
fsh_morphospecies	fsh_perFish	morphospeciesID
fsh_morphospecies	fsh_bulkCount	morphospeciesID
fsh_fieldData	fsh_morphospecies	Join not recommended: data can be related via date of sampling
fsh_perPass	fsh_morphospecies	Join not recommended: data can be related via date of sampling

Table 1	Table 2	Join by field(s)
fsh_bulkCount	fsh_fieldData	Requires intermediate table: join via fsh_perpass table
fsh_bulkCount	fsh_perFish	Join not recommended: data can be related via date of sampling
fsh_fieldData	fsh_perFish	Join not recommended: data can be related via date of sampling

Documentation

NEON Aquatic Sampling Strategy
NEON.DOC.001152vD | 846.5 KiB | PDF

AOS Protocol and Procedure: FSS – Fish Sampling in Wadeable Streams
NEON.DOC.001295vH | 2.5 MiB | PDF

AOS Protocol and Procedure: FSL – Fish Sampling in Lakes
NEON.DOC.001296vJ | 2.4 MiB | PDF

NEON Algorithm Theoretical Basis Document (ATBD): OS Data Quality Control
NEON.DOC.005424vA | 651.3 KiB | PDF

NEON User Guide to Fish electrofishing, gill netting, and fyke netting counts (NEON.DP1.20107)
NEON_fish_userGuide_vD | 556.3 KiB | PDF

NEON AOS Design Optimization: Fish sampling in Streams and Lakes
fsh_optimizationReport | 760.5 KiB | PDF

For more information on data product documentation, see:
https://data.neonscience.org/data-products/DP1.20107.001

Citation

To cite data from value: "Fish electrofishing, gill netting, and fyke netting counts" (DP1.20107.001), see citation here:
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